

Theory and manufacturing procedures for a cast nozzle

Theory of production methods

There are several manufacturing methods available:

- **turning** (the final product is gradually formed by removing material on a lathe, CNC machine, or similar device)
- **casting** (a homogeneous material is poured into a negative/mold, and the finished product is removed after hardening)
- **layering** (layers of material are applied to a negative, subsequently forming the finished product)

We decided to use casting, as it is a simple and cost-effective method for producing durable parts.

Tested and used materials

Ideal material

An ideal material is understood as one that possesses the ideal properties for a given application.

In our case, the ideal material is one that is lightweight and highly resistant to heat and pressure. It is also easy to process (cast).

Consideration

There are many castable materials available on the market. An ideal choice would be a material such as refractory concrete, which offers high heat resistance—up to 1,700°C. However, this material is expensive, sold only in inconvenient 25 kg bags, and is too viscous. Therefore, we considered using more accessible materials—specifically, a mixture of sand, gypsum, and concrete mix (hereinafter referred to simply as concrete).

Our reasoning:

1. **concrete mix** – provides strength and heat resistance
2. **plaster** - lightens the mixture and fills the intricate, fine geometry of the mold
3. **Sand** - it provides strength, as it acts as a medium for the concrete

Tested mixtures

We tested various ratios of the three previously mentioned components (hereinafter referred to as the "three-component mixture"). We aimed to achieve a ratio where the resulting measured properties would best meet the definition of an ideal material. The tested mixtures include:

variant	ratio [mass]	material
v1	13:26:7:40	plaster : Knauf_concrete : water : sand : water
v2	12:5	Knauf_concrete : water
v3	35:35:22	Knauf_concrete : sand : water
v4	25:4	Fofrbeton_concrete : water

Based on the tests conducted so far, we have concluded that V4 is an ideal mixture. Plaster was not used—contrary to the initial consideration—as it had an adverse effect on the material's final strength..

Equipment and Tooling

Mold Theory and Design

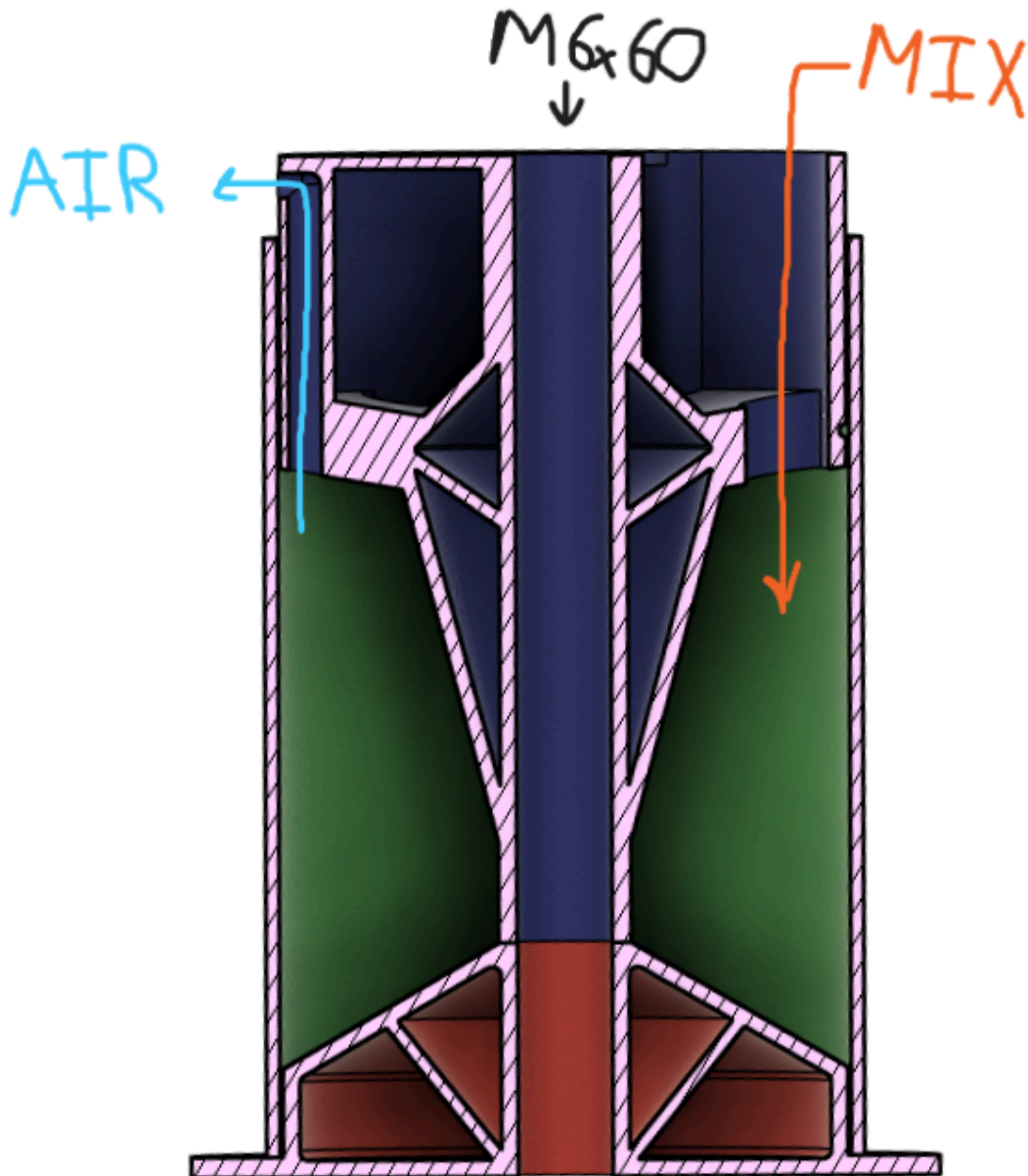
When designing a mold for casting a high-quality part, it is essential to keep the following in mind:

1. air vents
2. ease of pouring the mixture
3. ease of casting removal
4. surface finish of the mold and the casting

Following these rules, we design a mold with:

- vents that allow air to escape
- with openings large enough to pour in the raw mixture
- easy disassembly

Our mold consists of three parts: 1. the converging section (red), 2. the diverging section + neck (blue), and 3. the outer shell (green), as shown in the image.



We 3D-printed parts 1, 2, and 3. For part 3, we recommend using a non-stick material; otherwise, you should expect this part to be ruined after use.

Mold making

We printed the mold on an FDM 3D printer, opting for a vertical print orientation to simplify the printing process. Using other types of 3D printers, such as SLA, is certainly suitable—and perhaps even more sensible.

Mixture vibration (device)

We constructed the vibrating device using four springs, wooden boards, an AC voltage regulator, and a universal carbon-brush motor. Suitable fasteners were also used.

Release agent

This is a solution that facilitates the removal of the casting from the mold. In our case, we used a mixture of water, IPA, and solid soap. We saturated hot water with solid soap and added an equal part of IPA. The mixture is applied in 2–4 layers using a brush and left to dry.

Casting theory and process

Theory

During casting, we must pay attention to the correct sequence of operations. We must also be mindful of the time after which the material becomes too stiff to cast.

Procedure

Line the inside of the mold's outer shell with a single layer of standard office paper, securing it at both ends with clear adhesive tape. Apply a release agent to the converging and diverging sections. Join the sections using an M6x60 bolt. Slide the prepared outer shell into place.

Prepare approximately 80 g of the dry "fofr_beton" mix and the corresponding amount of water (approx. 13 g). Mix thoroughly just before casting.

Place the mold on a vibration table, set an appropriate vibration intensity, and gradually pour the mixture into the mold, taking care not to block the air vent channels. After filling, continue vibrating for at least another 1–2 minutes.

Leave the mold in a shaded area at a temperature of 15–25°C. For a stronger casting, remove the outer shell after 1–2 days of setting and submerge the mold containing the hardening material in water for approximately 7–14 days.

Warnings and recommendations

Warnings

1. The specific material used must be carefully selected and compared with other options
2. A calculation error could have adverse and drastic consequences.
3. The material's internal bonding degrades at high temperatures. This nozzle is intended for single use and for motors with short burn times.
4. As advertised, concrete reaches its maximum strength after about 28 days of hardening.

Recommendations

1. It is advisable to overfill the mold. The heavier material will settle, displacing the thin, low-quality material to the top.
2. Conducting the appropriate tests is a necessity rather than a recommendation.